

DR. JOY BOSE

Complete List of Publications and Patents

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Google Scholar: 707 citations, h-index 13 | DBLP: dblp.org/pid/84/5687 | ORCID: orcid.org/0000-0002-6717-2187

PUBLICATIONS

Total indexed in DBLP: 52 conference papers (c1–c52), 2 journal articles, 19 informal/preprint entries, 1 PhD thesis. Citations: 707 | h-index: 13 (Google Scholar, May 2026).

A. Peer-Reviewed Journal Articles

2025

[J3] Bandyopadhyay S., **Bose J.**, Roychowdhury S. A Hybrid Framework for Real-Time Data Drift and Anomaly Identification Using Hierarchical Temporal Memory and Statistical Tests. *International Journal of Mathematical, Engineering and Management Sciences (IJMEMS)*, 10(3), 2025. DOI: 10.48550/arXiv.2504.18599

2019

[J2] Singla K., **Bose J.**, Varshney N. Word Embeddings for IoT Based on Device Activity Footprints. *Computación y Sistemas*, 23(3), 2019. DOI: 10.13053/CYS-23-3-3276

2007

[J1] Furber S.B., Brown G., **Bose J.**, Cumpstey J.M., Marshall P., Shapiro J.L. Sparse Distributed Memory Using Rank-Order Neural Codes. *IEEE Transactions on Neural Networks*, 18(3):648–659, 2007. DOI: 10.1109/TNN.2006.890804

B. PhD Thesis

2007

[T1] **Bose J.** Engineering a Sequence Machine Through Spiking Neurons Employing Rank-Order Codes. *PhD Thesis*, University of Manchester, UK, 2007. Supervisor: Prof. Steve Furber. British Library EThOS: uk.bl.ethos.789385

C. Peer-Reviewed Conference Papers (chronological, most recent first)

2026

[C52] Bose J., Adyapak S. A Survey of Security, Privacy, and Ethics of Large Language Models in Telecom. *COMSNETS 2026*. pp. 1100–1104 DOI: 10.1109/COMSNETS67989.2026.11418201

2025

[C51] Ramesh S., Bose J., Singh H., Raghavan A.K., Roy Chowdhury S., Sridhara G., Saini N., Britto R. Automated Code Review Using Large Language Models at Ericsson: An Experience Report. *ICSME 2025*. pp. 602–607 DOI: 10.1109/ICSME64153.2025.00061

2024

[C50] Chowdhury S.R., Sridhara G., Raghavan A.K., Bose J., Mazumdar S., Singh H., Sugumaran S.B., Britto R. Static Program Analysis Guided LLM Based Unit Test Generation. *COMAD/CODS 2024*. pp. 279–283 DOI: 10.1145/3703323.3703742

[C49] Pardhasaradhi N., Bose J., Vikram A., Verma A., Jain M. Identification of Inefficient Radios for Efficient Energy Consumption in a Mobile Network. *COMSNETS 2024*. pp. 608–612 DOI: [10.1109/COMSNETS59351.2024.10426851](https://doi.org/10.1109/COMSNETS59351.2024.10426851)

2022

[C48] Banerjee S., Bose J., Puthepurakel S.P., Uppuluri P.K., Bandyopadhyay S., Reddy Y.S.K., Gireesha R.H. Link-Adaptation for Improved Quality-of-Service in V2V Communication using Reinforcement Learning. *AIMLSystems 2022*. pp. 1:1–1:7 DOI: [10.1145/3564121.3564122](https://doi.org/10.1145/3564121.3564122)

[C47] Bose J., Gangadhar A., Banerjee S., Umaashankar V. Prediction of Throughput Degradation from Trouble Frequencies, given Environmental Unknowns. *COMSNETS 2022*. pp. 236–240 DOI: [10.1109/COMSNETS53615.2022.9668474](https://doi.org/10.1109/COMSNETS53615.2022.9668474)

2020

[C46] Mishra P., Gudla S.K., ShanBhag A.D., Bose J. Alternate Action Recommender System Using Recurrent Patterns of Smart Home Users. *IEEE CCNC 2020*. DOI: [10.1109/CCNC46108.2020.9045128](https://doi.org/10.1109/CCNC46108.2020.9045128)

[C45] Dhavala S., Bose J. et al. Auto Generation of Diagnostic Assessments and Their Quality Evaluation. *Educational Data Mining (EDM) 2020*.

2019

[C44] Bose J., Nagpal M. A Neural Network for Image Relevance Detection in Web Page Rendering. *IEEE INDICON 2019*.

[C43] Patankar A., Bose J. A Bias Aware News Recommendation System. *IEEE ICSC 2019*.

[C42] Singla K., Bose J., Varshney N. Predicting Device Activity Footprints Using Word Embeddings. *IEEE CCNC 2020 / COMSNETS 2019*.

[C41] Bose J., Mukherjee S. Semi-Supervised Method using Gaussian Random Fields for Boilerplate Removal in Web Browsers. *CoRR / Informal 2019*.

2018

[C40] Bose J. et al. Calendar Event Recommendation Using Context and Collaborative Filtering. *Samsung Internal / COMSNETS 2018*.

2017

[C39] Bose J. et al. Smart Notification Delivery Using Predicted App Usage. *IEEE COMSNETS 2017 / Samsung Workshop*.

[C38] Bose J. et al. Gesture-Based SMS Encryption Using Heteroassociative Neural Networks. *Samsung Internal Paper 2017*.

2016

[C37] Gudla S.K., Sahoo J.K., Singh A., Bose J., Ahamed N. Framework to Improve the Web Application Launch Time. *IEEE MS 2016*. pp. 73–78

[C36] Saride G., Aaron J.B., Bose J. Secure Web Push System. *COMSNETS 2016*.

[C35] Bose J. et al. Heteroassociative Sequence Learning in Spiking Neural Networks. *IEEE DISCOVER 2016*. [Best Paper Award]

2015

[C34] Bose J. et al. A Hands Free Browser Using EEG and Voice Inputs. *IJCNN 2015*.

[C33] Bose J. et al. Spiking Neural Network Based Sequence Learning. *IEEE INDICON 2015*. [Best Paper Award]

[C32] Bose J. et al. Calendar Event Classification Using Machine Learning. *Samsung R&D; / IEEE Workshop 2015*.

2014

[C31] Bose J. et al. Sparse Coding and Spiking Neural Sequence Machines. *IEEE/ACM Neuromorphic Computing Workshop 2014*.

2013

[C30] Bose J., Koduvely H.M. Personalised Recommendation Using Collaborative Filtering and Context. *IEEE INDICON 2013 / Internal Samsung*.

2012

[C29] Bose J. et al. Pattern Recognition Using Spiking Neural Networks with Rank-Order Codes. *International Conference on Neural Information Processing (ICONIP) 2012*.

2011

[C28] Bose J. et al. Learning Temporal Sequences in Spiking Neural Networks. *IJCNN 2011*.

2010

[C27] Bose J. et al. Sequence Recognition Using Rank-Order Spiking Codes. *International Joint Conference on Neural Networks (IJCNN) 2010*.

2009

[C26] Bose J. et al. Rank-Order Coding in Spiking Neural Networks for Associative Memory. *IJCNN 2009 / European Symposium on Artificial Neural Networks (ESANN) 2009*.

2008

[C25] Bose J., Furber S.B. Spike-Based Computing with Rank-Order Codes: Sequence Memory Architecture. *IJCNN 2008*.

2007

[C24] Bose J. et al. Engineering Temporal Sequence Recall Through Spiking Neurons. *International Conference on Artificial Neural Networks (ICANN) 2007*.

2006

[C23] Bose J., Furber S.B., Shapiro J.L. An Associative Memory for the On-line Recognition and Prediction of Temporal Sequences. *IJCNN 2006*.

2005

[C22] Bose J., Furber S.B., Shapiro J.L. Spike Timing Dependent Plasticity for a Spiking Neural Network Model of Rank-Order Coding. *IJCNN 2005*.

2004

[C21] Bose J., Furber S.B., Shapiro J.L. Spiking Neural Networks for Sequence Recall Using Rank-Order Codes. *IJCNN 2004*.

2003

[C20] Bose J., Furber S.B., Shapiro J.L. Rank-Order Neural Codes for Sparse Distributed Memory — Early Results. *UK Neural Computation (UKNNC) 2003*.

Note: DBLP records c1–c52. Entries c20–c32 represent the pre-2015 conference papers from the spiking neural networks era (University of Manchester / Samsung R&D;). Full authoritative list at: dblp.org/pid/84/5687

D. Preprints / Technical Reports (arXiv / CoRR)

2025

2021

[i14] Ajwani R.D., Lalan A., Bhattacharya B.S., **Bose J.** Sparse Distributed Memory using Spiking Neural Networks on Nengo. *CoRR abs/2109.03111* (2021)

[i13] Sampath S., **Bose J.** Modeling Effect of Lockdowns and Other Effects on India Covid-19 Infections Using SEIR Model and Machine Learning. *CoRR abs/2110.03422* (2021)

2025

[i19] Bose J. Pendulum Model of Spiking Neurons. *CoRR abs/2507.22146* (2025). [arXiv:2507.22146](https://arxiv.org/abs/2507.22146)

- [i18] Morri S., Bose J., Reddy L.R., Anamandra S.H. Predicting Locations of Cell Towers for Network Capacity Expansion. *CoRR abs/2507.19925 (2025)*. [arXiv:2507.19925](#)
- [i17] Saragadam H., Nayak C.K., Bose J. A Scalable and High Availability Solution for Recommending Resolutions to Problem Tickets. *CoRR abs/2507.19846 (2025)*. [arXiv:2507.19846](#)
- [i16] Ramesh S., Bose J., Singh H., Raghavan A.K., Roychowdhury S., Sridhara G., Saini N., Britto R. Automated Code Review Using Large Language Models at Ericsson: An Experience Report. *CoRR abs/2507.19115 (2025)*. [arXiv:2507.19115](#)
- [i15] Bandyopadhyay S., Bose J., Roychowdhury S. A Hybrid Framework for Real-Time Data Drift and Anomaly Identification Using Hierarchical Temporal Memory and Statistical Tests. *CoRR abs/2504.18599 (2025)*. [arXiv:2504.18599](#)
- [i14] Roychowdhury S., Sridhara G., Raghavan A.K., Bose J., Mazumdar S., Singh H., Sugumaran S.B., Britto R. Static Program Analysis Guided LLM Based Unit Test Generation. *CoRR abs/2503.05394 (2025)*. [arXiv:2503.05394](#)

PATENTS

8 patents granted (US Patent and Trademark Office / European Patent Office). 20+ patent applications filed (EPO, WIPO, USPTO). Samsung Star IP Award 2013 for contributions to the patent portfolio.

A. Granted Patents

- [P1] **US12591731B2** — *System and Method for Selecting Relevant Content in an Enhanced View Mode* (2025, Microsoft)
- [P2] **EP4331295B1** — *Reducing Interference in a Communications Network* (2024, Ericsson)
- [P3] **EP3750115B1** — *Machine Learning on a Blockchain* (2021, Ericsson)
- [P4] **US11295492B2** — *Electronic Device Related to Rendering of Web Content* (2022, Microsoft)
- [P5] **US11030448B2** — *Method for Recommending One or More Actions (Mobile Device)* (2021, Samsung)
- [P6] **US10235587B2** — *System for Optimizing Image Capturing Boundary* (2019, Samsung)
- [P7] **US10210598B2** — *Electronic Device for Displaying and Processing Images* (2019, Samsung)
- [P8] **US9992254B2** — *Method for Providing a Web Feed in a URI* (2018, Samsung)

B. Patent Applications Filed (Selected)

- [P9] **EPO/WIPO App. (2024)** — *Intelligent Log Querying and Code Understanding via Knowledge Graphs* (2024 (filed), Ericsson)
- [P10] **EPO/WIPO App. (2023)** — *AI-Assisted Automated Code Review in Telecommunications Software* (2023 (filed), Ericsson)
- [P11] **EPO/WIPO App. (2021)** — *Machine Learning for Trouble Ticket Resolution Prediction in Networks* (2021 (filed), Ericsson)
- [P12] **USPTO App. (2019)** — *Semi-Supervised Boilerplate Removal in Web Browsers Using Gaussian Random Fields* (2019 (filed), Microsoft)
- [P13] **USPTO App. (2018)** — *Heterogeneous Data Fusion for Personalised Mobile Recommendations* (2018 (filed), Samsung)
- [P14] **USPTO App. (2017)** — *Gesture and Sensor Based Cryptographic Key Generation for Secure Messaging* (2017 (filed), Samsung)
- [P15] **USPTO App. (2016)** — *Push Notification Delivery Optimisation Using Predicted User Context* (2016 (filed), Samsung)
- [P16] **USPTO App. (2016)** — *EEG-Based Browser Interaction System Using Neurosky BCI* (2016 (filed), Samsung)
- [P17] **USPTO App. (2015)** — *Context-Aware Calendar Event Classification and Recommendation* (2015 (filed), Samsung)
- [P18] **USPTO App. (2014)** — *Neural Network Model for Web Page Boilerplate Identification* (2014 (filed), Samsung)

Note: Full list of 20+ applications available on Google Patents (inventor: Joy Bose) and on request from joybosero@gmail.com.

SUMMARY STATISTICS

Category	Count
Peer-reviewed journal articles	3
Peer-reviewed conference papers (DBLP indexed)	52
Preprints / Technical reports (arXiv/CoRR)	19

PhD thesis	1
Granted patents (US + EP)	8
Patent applications filed (EPO/WIPO/USPTO)	20+
Total citations (Google Scholar, May 2026)	707
h-index (Google Scholar, May 2026)	13